

Shaine Ismail

Head of Data & AI Architecture | Enterprise Technology Leadership | Financial Services

linkedin.com/in/shaineismail

25+ years in senior data, architecture & engineering roles across financial services

Areas

- Technology Leadership
- Team & Squad Leadership
- Executive Engagement
- Data Architecture
- Cloud Platforms
- Big Data & Streaming
- Financial Crime
- Cyber Security
- Machine Learning
- Technical Governance

Clients

HSBC, NatWest / RBS, Deutsche Bank, JP Morgan, Swiss Re, Lehman Brothers, Royal Bank of Canada

Qualifications

MSc Information Technology, Stirling University
MA (Hons) Economics, Heriot Watt University

Twenty-plus years setting technology and data strategy, and leading the architecture and engineering functions, behind some of the UK’s largest financial institutions.

- **Led enterprise data and architecture strategy** across major transformation programmes — setting direction, owning design authority, and leading delivery teams and squads of 100+ people across multiple workstreams, suppliers and geographies.
- **Combines hands-on architecture leadership with board-level commercial responsibility** — co-founded and helped scale a specialist consultancy to multi-million-pound revenue, shaping strategy, client relationships and recruitment alongside senior client-side delivery.
- Domain depth in **Financial Crime, Data Platforms and Cloud**, built across **Royal Bank of Canada, Swiss Re, Lehman Brothers, JP Morgan, Deutsche Bank, RBS/NatWest and HSBC.**

Career chronology 2000 – 2026

2020

Bigspark
Co-Founder & Director

Mar 2020 – Current

Co-founded and scaled Bigspark, a specialist data and software engineering consultancy focused primarily on financial services. Combined board-level

responsibility for the growth and direction of the company with senior architecture and engineering leadership for major clients.

- Co-led the growth of Bigspark from an early-stage consultancy into a ~£8m-revenue, 75-person technology business serving banking, insurance and asset-management clients
- Shaped the firm's technology and data propositions, particularly across data platforms, cloud, engineering, analytics and AI
- Led Bigspark's strategic technology partnerships with AWS, Databricks, Snowflake, Anthropic and TD SYNEX, embedding their platforms across client delivery
- Co-owned company strategy, business development, key client relationships, recruitment and the development of engineering capability
- Advised a portfolio of clients beyond financial services on data and AI strategy, including a technology supplier to the Home Office
- Continued to operate directly in senior client roles, including Principal Engineer, Technical Lead and Head of Data Architecture engagements
- Partnered with senior executives and technology leaders, translating business strategy into architecture, delivery roadmaps and scalable engineering capability

2026 **HSBC**

Mar – Aug 2026

Head of Asset Management Data Architecture

Engaged to relaunch the Design Authority for HSBC Asset Management's Data Architecture function, setting target-state direction and governance.

- Relunched the Design Authority for Data Architecture, re-establishing governance and target-state direction for the Asset Management business
- Oversaw AI-enabled business process, data product and data quality work across the function
- Led a Data Architecture & Governance team of 4, embedded within a ~£2–3m data programme delivered jointly with the Head of Data for Asset Management

2025 **NatWest Retail**

Sep 2025 – Feb 2026

Architecture Consultant, CORA

Led Bigspark's architecture and DevOps engagement for NatWest's CORA programme, spanning three flagship AI capabilities — Real Time Voice, Mortgage Assistant and Fraud Analyser.

- Led design of NatWest's first real-time voice AI use case, built on OpenAI

2024 **NatWest Group**

Jun 2024 – Jun 2025

Technical Lead, Data Cloud X — Data Marketplace

A data transformation programme developing the operate and technology capability to enable the bank's migration to a federated data product model.

- Directed technical delivery across all 5 workstreams, with 100+ FTEs spanning the UK and India

- Initially engaged to recover the programme — established the product roadmap, reset the delivery plan, and aligned senior stakeholders
- Delivered the first wave of data products and all data management controls into production
- Managed supplier relationships with Accenture and AWS
- Sat on the Data Platforms Technology Leadership team, shaping platform-wide technical direction

2021 **NatWest Group**

Jun 2021 – Jun 2024

Principal Engineer, Financial Crime Hub — pKYC

Responsible for architecture and technical delivery across Data Science & Analytics, Detection (Quantexa), the Analytics Mirror (Hubble), and MI and Reporting — four platform domains within NatWest's Financial Crime Hub.

- Held end-to-end platform design authority: identified gaps in existing designs, then set direction for platform capability into 2025/26 — including customer multi-relationship capability and the Client, Pooled Monies and Trade Finance product onboarding
- Led technical delivery of the Policy Rules Engines
- Took the Reporting and Analytics Mirror, a Hadoop & Tableau platform, from design through to delivery
- Governed the architecture for Quantexa's migration to AWS
- Directed delivery of the FinCrime Analytics and MI Hub for CDD — leading 3 squads of engineers, DevOps, data scientists and reporting specialists through a full lake layered-house design and transition into BAU operations

2018 **Royal Bank of Scotland**

Jul 2018 – Jun 2021

Principal Engineer, Data Engineering

Set engineering direction across RBS's Data Domain — Big Data, Streaming and Data Science.

- Steered the product roadmap for all streaming, big data and Machine Learning platforms across RBS Data
- Shaped technical strategy for Payments Transformation (streaming, off-premise) and Data Cloud enablement (big data and data science capability)
- Stood up a strategic on-premise Data Science Platform, built on open source technologies
- Delivered the RBS Event Streaming Platform, including real-time data transformation with Spark
- Recruited and led a team delivering a bank-wide real-time Log Capture service for the RBS Cyber Security Programme
- Drove RBS's technology adoption of Apache Kafka, Apache Flume and StreamSets to support real-time platforms

2015

Royal Bank of Scotland

Jun 2015 – Jul 2018

Architecture & Engineering Lead, Data Platforms

Engaged by Royal Bank of Scotland to lead the Big Data Engineering rollout of Hadoop as a platform for the bank.

- Stood up Hadoop security via Active Directory integration, subsequently adopted bank-wide by Group Unix
- Set up an external audit Software Currency process across Analytics Platforms
- Pioneered cloud-based software engineering across Data Platforms
- Advised on strategic infrastructure investment across Big Data and Analytics
- Drove technical adoption of all new Big Data and Analytics tooling bank-wide

– via SNI Services Limited, own contracting entity, Mar 2008 – Feb 2020 –

2013 Deutsche Bank

Oct 2013 – Apr 2015

Hadoop Solution Architect, DataLab Store

Built Deutsche Bank's Process and Analytics Lab data hub; led an offshore team delivering 300+ ETL feeds in three months and acted as Hadoop SME for the group.

2012 Top Right Group

Oct 2012 – Oct 2013

Lead Developer, Big Data

Built a quantitative analytics tool, [wgsninstock.com](http://www.wgsninstock.com) <http://www.wgsninstock.com>, for WGSN — data workflow, MapReduce pipeline and authentication layer end-to-end.

2011 Swiss Re

Aug 2011 – Aug 2012

Technical Governance & Design, Risk Reporting System

Overhauled the risk-reporting development environment — agile release cycle, TDD (14,000+ tests) and CI — while building the Solvency II Market Risk feed.

2010 Royal Bank of Canada

Aug 2010 – Jul 2011

Senior Developer, Greenfield Market Risk System

Built a Market Risk Aggregation Warehouse in Java, with real-time aggregation (~600ms/trade) and golden-source persistence that cut load time by 70%.

2008 Swiss Re

Mar 2008 – Aug 2010

Technical Lead, Product Control

Ran a strategic daily P&L reporting platform with a global user base, working directly with the Global Head of Product Control on Swiss Re's Risk Budgeting and the BlackRock outsourcing.

– earlier career, permanent roles –

- 2005

Lehman Brothers

Dec 2005 – Mar 2008

Lead Developer, CAD II / Market Risk System

Built trading systems (Java, Perl, Sybase, VB) for FI, Equities & Commodities, including reconciliation for 20M+ trades/day.
- 2001

JP Morgan & JP Morgan Asset Management

Oct 2001 – Dec 2005

Senior Developer, Fixed Income Research

Graduate role on Fixed Income and Structured Equity Applications, moving into FI Markets Research development and analytics for benchmark spreads and index-linked bonds.
- 2000

British Telecom

Jan 2000 – Oct 2001

Marketing Manager, BT Internet

Built MIS systems and reported on consumer products for BT Internet.

Board & Entrepreneurial Experience

CTO · Jan 2015 – Sep 2021

Carsift Limited

Directed product and engineering as CTO, building and leading the team behind a machine learning car-recommendation platform; winner, BNP Startup Bootcamp 2016. [carsift.co](http://www.carsift.co/) <http://www.carsift.co/>

Board Member · Jul 2023 – current

Gighabit Limited

Streamlined ticketing platform for fans and event organisers.

Board Member · Sep 2022 – Apr 2024

My Trauma Therapy Limited

Support and safety net for victims of narcissistic abuse. mynara.app <https://mynara.app/>

CTO / Trustee (voluntary) · Apr 2012 – Apr 2017

Kubaru Microfinance

Led technology end-to-end for a microfinance charity — £15k of loans to 50+ entrepreneurs across 3 continents.

Capabilities

Enterprise Data Architecture	Target-state design, data platform strategy, federated data product models
Data & AI Strategy	Cloud data strategy, AI/ML enablement, technology roadmaps
Data Products	Data marketplace delivery, product onboarding, data management controls

Cloud & Platform Architecture	AWS/Azure platform design, Hadoop-based data platforms, real-time streaming
Architecture Governance	Design authority, technical standards, investment prioritisation, technology risk
Engineering Leadership	Squad & team leadership (100+ FTEs), recruitment, capability building
Financial Crime & Risk	KYC platforms, Quantexa, detection & analytics architecture
Executive & Board Engagement	Board, C-suite and strategic partner relationships (Accenture, AWS, Databricks, Snowflake, Anthropic, TD SYNEX)

Technologies

AWS · Azure · Databricks · Snowflake · Anthropic · Hadoop · Spark · Kafka · Quantexa
· Tableau · Java · Python